

Jiaqi Tang

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

Bachelor of Science in **Physics** in the Grainger College of Engineering

Expected May 2027

Bachelor of Science in **Mathematics** in the College of Liberal Arts & Sciences

Expected May 2027

-**Cumulative GPA:** 3.90/4.00; James Scholar Candidate, Dean's List for 2024-2025 academic years, Illinois Engineering Outstanding Scholarship, Illinois Engineering Achievement Scholarship.

-**Relevant Coursework:** Stochastic Processes, Probability Theory, Numerical Methods/Analysis, Linear Algebra, Differential Equations, Real Analysis, Data Structures, Quantum Information/Mechanics.

RESEARCH EXPERIENCE

Illinois Risk Lab

Champaign, IL

Undergraduate Researcher

Jan 2026 - Present, On-site

-Conducting research under the supervision of Prof. Xiaochen Jing, focusing on Maximal Extractable Value (MEV) and the detection of sandwich attacks on the Ethereum network.

-Developing a Machine Learning-based detection framework utilizing LSTM for temporal pattern recognition and Graph Neural Networks (GNN) to identify predatory trading behaviors from real-time mempool data and on-chain transaction history.

-Engineering features based on gas priority fee bidding and transaction timing gaps to quantify value extracted from users, while evaluating mitigation strategies such as Smart RPC screening and private transaction routing.

Illinois Mathematics Lab

Champaign, IL

Undergraduate Researcher

Jan 2025 - May 2025, On-site

-Worked under the supervision of Prof. Daniel Cooney, focusing on Stochastic Models of Cross-Scale Evolutionary Dynamics. I worked on developing systems of ODEs/PDEs from Markov Processes and mutation dynamics, and analyzed the ODE stability in the continuous limit. I employed the Euler method and the RK4 method for numerical solutions to analyze the steady-state convergence characteristics of dynamical models.

PROFESSIONAL EXPERIENCE

Huafu Securities Co., Ltd

Shanghai, China

Research Assistant Financial Engineering Internship

May 2025 - Aug 2025, On-site

-Engineered financial features (e.g., Herfindahl-Hirschman Index, industry drift, benchmark deviation) for 5,000+ Chinese open-ended funds to quantify portfolio concentration and thematic bias relative to the CSI 300.

-Applied machine learning methods (hierarchical clustering and Gaussian Mixture Models (GMM)) to group funds and select several funds per cluster for inclusion in an Enhanced Portfolio Optimization (EPO) strategy.

-Built a Bayesian state-space model combining a Kalman filter to infer daily fund holdings from quarterly disclosures. Achieved NAV reconstruction with a MAE less than 0.01 across an entire quarter, demonstrating a scalable framework for intraday portfolio inference and alpha attribution.

University of Illinois Urbana-Champaign – The Grainger College of Engineering

Champaign, IL

Expert Learning Assistant for PHYS 212 (Electricity & Magnetism) /Peer Mentor Aug 2024 - May 2025, On-site

-Prepared experiment materials, assisted with lab design, and data analysis, and provided hands-on guidance to students during experiments. Identified student challenges and provided feedback to instructors.

-Supporting newly admitted students in their transition to studying Physics at UIUC.

SELECTED PROJECTS

LLM-Driven Multimodal Quantitative Trading System (A-Share)

<https://github.com/TanhJK728/A-Share-LLM-Trading-Research>

Dec 2025-Present

-Architected an end-to-end quantitative trading pipeline fusing GBDT factors with LLM-driven sentiment analysis (RAG), utilizing MongoDB to parse unstructured financial news and ClickHouse for OLAP factor storage.

- Engineered a dynamic prompt engineering module using Jieba for entity extraction, quantifying news impact relative to real-time market cap contexts to filter noise.
- Backtested the multi-factor strategy via Microsoft Qlib, achieving a Sharpe Ratio of 4.51 and significantly outperforming the CSI 300 benchmark by capturing cross-sectional alpha.

Short-Term “Washout & Burst” Strategy & Factor Mining (A-Share)

<https://github.com/TanhJK728/A-Share-Washout-Research>

Dec 2025-Present

- Developed a short-term trading strategy using LightGBM to model price-volume microstructure (e.g., lower shadow ratio, volume shrinkage), identifying smart money flow signals from daily OHLCV data.
- Built a robust data ingestion pipeline using AkShare and ClickHouse to process full-market historical data, enabling efficient learning-to-rank model training.
- Optimized strategy execution with a dynamic volatility filter, achieving a Sharpe Ratio of 4.66 and 1284% annualized return during bull market phases in out-of-sample tests.

AWARDS AND ACHIEVEMENTS

- National Silver Medal in the 34th Spanish Physics Olympiad Apr 2023
- American Invitational Mathematics Examination Qualifier (Scored 13/15 in AIME I) Feb 2023

SKILLS

- Programming Languages: Python (numpy, pandas, scikit-learn, PyTorch), MATLAB, LaTeX
- Language: Mandarin (Native), Spanish (Native), English (Native), Catalan (Advanced)

COMMUNITY SERVICE

2024-2025 Urbana-Champaign Senate Membership